INCH-POUND
MIL-DTL-32624
23 January 2019
SUPERSEDING
MC/PS 04 2015 SYSCOM
15 March 2015

DETAIL SPECIFICATION

DRAWERS, FLAME RESISTANT, MIDWEIGHT, COLD WEATHER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flame resistant (FR), midweight, cold weather drawers (see 6.1). The drawers will be one type and in the following sizes and lengths, as specified (see 6.2).

Extra-Small	Small	Medium	Large	Extra-Large
Regular	Regular	Regular	Regular	Regular
		Long	Long	Long

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-55195 - Thread, Para-Aramid, Spun, Intermediate Modulus

A-A-55217 - Thread, Aramid, Spun Staple

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8415

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9	-	Visual Assessment of Color Difference of Textiles
AATCC TM8	-	Colorfastness to Crocking: Crockmeter Method
AATCC TM16.3	-	Colorfastness to Light: Xenon-Arc
AATCC TM20A	-	Fiber Analysis: Quantitative
AATCC TM61	-	Colorfastness to Laundering: Accelerated
AATCC TM135	-	Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D76/D76M	-	Standard Specification for Tensile Testing Machines for Textiles
ASTM D737	-	Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1577	-	Standard Test Methods for Linear Density of Textile Fibers
ASTM D1776/D1776M	-	Standard Practice for Conditioning and Testing Textiles
ASTM D1777	-	Standard Test Method for Thickness of Textile Materials
ASTM D2594	-	Standard Test Method for Stretch Properties of Knitted Fabrics Having Low Power
ASTM D3512/D3512M	-	Standard Test Method for Pilling Resistance and Other Related Surface Changes of Textile Fabrics: Random Tumble Pilling Tester
ASTM D3774	-	Standard Test Method for Width of Textile Fabric
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3787	-	Standard Test Method for Bursting Strength of Textiles–Constant-Rate-of-Traverse (CRT) Ball Burst Test
ASTM D5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D5103	-	Standard Test Method for Length and Length Distribution of Manufactured Staple Fibers (Single-Fiber Test)
ASTM D6193	-	Standard Practice for Stitches and Seams
ASTM D6413/D6413M	-	Standard Test Method for Flame Resistance of Textiles (Vertical Test)

(Copies of these documents are available online at www.astm.org.)

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CRC PRESS

Hayes, A. Wallace (editor) and Kruger, Claire L. (editor). *Hayes' Principles and Methods of Toxicology*. 1989. 394-396.

(There are a number of ways through which this book may be obtained, such as online retailers, bookstores, public library, to name a few.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 11092 - Textiles – Physiological Effects – Measurement of Thermal and Water-Vapour Resistance Under Steady-State Conditions (Sweating Guarded-Hotplate Test)

(Copies of this document are available online at www.iso.org.)

JOHN WILEY & SONS INC

Marzulli, Francis and Maibach, Howard. "Contact Allergy: Predictive Testing in Humans." *Advances in Modern Toxicology: Dermatoxicology and Pharmacology, Volume 4*. John Wiley & Sons Inc. 1977. 353-372.

(Copies of this document are available from the U.S. Army Center for Health Promotion and Preventative Medicine, ATTN: MCHB-DC-TTE, Bldg., E-2100, Aberdeen Proving Ground, MD 21010-5422.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to a conformance inspection in accordance with 4.3.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials and components.

3.3.1 Materials.

3.3.1.1 Basic material. The basic material for the drawers shall be a FR plaited knit fabric, constructed with a modacrylic/polyester/rayon/spandex blend, Polartec style 91371, or equal. The color of the fabric shall be Coyote 498 matching the reference standard furnished by the Government (see 6.2). The fabric shall show no toxicity when used as intended (see 6.1) and when tested as specified in 4.4.5. The fabric shall conform to the physical requirements specified in table I when tested as specified in 4.4.1.

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TABLE I. Basic material physical requirements.

Material Characteristic	Material Requirement
Construction	Plaited knit with face to back yarn denier gradient 1:2 differential
Fiber content	Modacrylic/polyester/rayon/spandex blend
Weight	7.625±0.625 ounces per square yard
Air permeability (minimum)	120 cubic feet/square feet/minute
Burst strength (minimum)	40 pounds
Colorfastness (minimum)	
Laundering (3 cycles)	4.0
Crocking	
Wet	3.0
Dry	4.0
Light, Xenon	4.0
Dimensional stability (3 cycles) (maximum)	
Wales	5.0%
Courses	5.0%
Fabric stretch, (minimum)	
Wales	35%
Courses	60%
Pilling (jersey face) (minimum)	
Initial	4.0
After laundering (3 cycles)	4.0
Thermal insulation (Clo) (minimum)	
Initial	0.3
After launderings (3 cycles)	0.25
Thickness	0.06 to 0.08 inch at 0.06 pounds per square inch
Vertical flame – initial	
Wales and courses (maximum)	
After flame	2.0 seconds
After glow	0 seconds
Char length	5.0 inches
Melt/drip	No melt/no drip
Vertical flame – after 5 launderings (maximum)	
Wales and courses	
After flame	2.0 seconds
After glow	0 seconds
Char length	5.0 inches
Melt/drip	No melt/no drip
Visual shade matching	
Daylight D75 (7500±200K)	Good match to reference standard
Incandescent (2856±200K)	Good match to reference standard

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3.3.1.2 Fly lining material. The fly lining for the drawers shall be a FR jersey knit fabric constructed with a modacrylic/polyester/rayon/spandex blend, Polartec style 2010, or equal. The color of the fabric shall be Coyote 498 matching the reference standard furnished by the Government (see 6.2). The fabric shall show no toxicity when used as intended (see 6.1) and when tested as specified in 4.4.5. The fabric shall conform to the physical requirements specified in table II when tested as specified in 4.4.1.

TABLE II. Fly lining physical requirements

Material Characteristic	Material Requirement
Construction	Jersey knit
Fiber content	Modacrylic/polyester/rayon/spandex blend
Weight	6.75 to 8.25 ounces per square yard
Bursting strength (minimum)	40 pounds
Colorfastness (minimum)	
Laundering (3 cycles)	4.0
Crocking	
Wet	3.0
Dry	4.0
Dimensional stability (3 cycles) - (maximum)	
Wales	5%
Courses	5%
Fabric stretch (minimum)	
Wales	30%
Courses	50%
Vertical flame – initial	
Wales and courses (maximum)	
After flame	2.0 seconds
Char length	5.0 inches
Melt/drip	No melt/no drip
Vertical flame – after 5 launderings	
Wales and courses (maximum)	
After flame	2.0 seconds
Char length	5.0 inches
Melt/drip	No melt/no drip
Visual shade matching	
Daylight (D75) (7500±200K)	Good match to reference standard
Incandescent (2856±200K)	Good match to reference standard
Water sorption/wicking	
Back and face	
Before laundering	Pass
After 20 launderings	Pass

3.3.2 Components.

3.3.2.1 Thread. The thread for the drawers shall be aramid, Tex Size 40 bobbin/looper conforming to A-A-55217. As an alternate, a para-aramid spun staple thread, Tex Size 39 conforming to A-A-55195, type I, may be used. The color of the thread shall be Coyote 498.

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3.3.2.2 Webbing, elastic with integrated draw-cord. The elastic webbing with the integrated draw-cord for the drawer waistband shall be a knit constructed from 1/150 semi dull (SD) textured polyester and a 34 gauge extruded latex elastic element. The color of the webbing with integrated draw-cord shall be Coyote 498. The webbing with integrated draw-cord shall conform to the physical requirements in table III when tested as specified in 4.4.1.

TABLE III. Webbing physical requirements.

Characteristic	Material Requirement
Fiber content	65/35 polyester/elastic
Width	1¼±1/16 inches
Weight (minimum)	0.55 ounce per linear yard
Elastic ends (minimum)	16
Picks per inch (minimum)	40
Warp ends (minimum)	16
Load to produce 50% elongation	0.8 to 4.5 pounds
Drift (maximum)	20%
Colorfastness	Equal to the basic material

3.3.2.2.1 Braided draw-cord. The braid for the tubular draw-cord shall be a knit constructed from 1/150 SD textured polyester in a continuous solid construction. The braided draw-cord shall be embedded in the elastic webbing of the waistband and the draw-cord shall be offset from the center. The draw-cord shall exit through the center front of the waistband on the finished drawers and shall be knotted. The color shall be Coyote 498. The draw-cord shall conform to the physical requirements in table IV when tested as specified in 4.4.1.

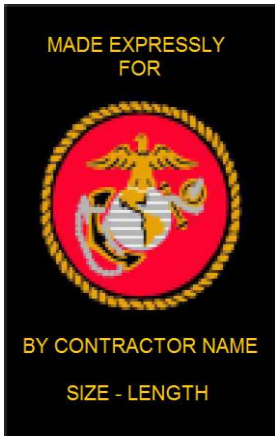
TABLE IV. Braided draw-cord requirements.

Characteristic	Requirement
Fiber identification	SD textured polyester
Construction	knit
Diameter	¼ (±1/32) inch
Colorfastness	Equal to the basic material

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3.3.2.3 Labels.

3.3.2.3.1 United States Marine Corps (USMC) exclusive label. Each pair of drawers shall have a woven USMC exclusive label, cut single, fused edge, 33 millimeters x 50 millimeters (1.299 inches x 1.969 inches), Bell Label, or equal. The exclusive label shall be placed inside of the drawers at the center back of the waist elastic and stitched on all four sides. The information on the label shall include the following: "Made Expressly for" at the top; followed by a woven USMC emblem in the center; the contractor's name below the emblem, and the abbreviated size – length designation at the bottom of the label in the configuration below. All lettering shall be in gold.



3.3.2.3.2 Flame resistant organizational gear (FROG) label. Each pair of drawers shall have a cut single, fused edge woven FROG label, 63 millimeters x 63 millimeters (2.480 inches x 2.480 inches) in size, manufactured by IbisTek, Bell Label, or equal. The FROG label shall be sewn on the outside of each pair of drawers to the left of the fly (as worn) ¾±¼ inch below the elastic waistband. The label shall be in the following configuration:



3.3.2.3.3 Combination size, identification, and instruction label. Each pair of drawers shall have a combination identification and instruction label conforming to type VI, class 14 of MIL-DTL-32075 when tested as specified in 4.4.1. The color of the label shall be white. The label shall be caught in the lower edge stitching of the USMC exclusive label. The combination label shall be facing the body and shall not be visible from the outside when the drawers are in use. The combination size, identification, and instruction label shall contain the following information and be in the following configuration:

Size and identification information:

DRAWERS, FR MIDWEIGHT
CONTRACT NO.:
NSN:
FIBER CONTENT:
CONTRACTOR'S NAME:
SIZE:

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Instruction Information:

Machine wash in lukewarm water (105 °F)

Tumble dry low or line dry

Do Not Use Fabric Softeners

DO NOT BLEACH, DRY CLEAN, OR IRON

3.3.2.3.4 Barcode label/tag. Each pair of drawers shall be individually barcoded with a label/tag conforming to type VI, class 17 of MIL-DTL-32075 when tested as specified in 4.4.1. The barcode label/tag shall be located so that it is completely visible on the item when it is folded or packaged as specified (see 6.2) and shall cause no damage to the item.

3.4 Design. The drawers shall have an elastic webbing waistband with an integrated draw-cord exiting at the center front, an access fly, and no cuff (see figures 1 and 2). The fly shall be backed with a lighter-weight knit material. All joining seams shall be flat locked construction to reduce chaffing and provide comfort in movement.

3.4.1 Figures. Figures 1 and 2 are furnished for informational purposes only. To the extent an inconsistency exists between the written specification and the figures, the written specification shall govern.

3.5 Patterns. Standard patterns, providing a seam allowance of ⅜ inch for all seams, will be furnished by the Government (see 6.2). The Government patterns shall not be altered in any way, and shall only be used for cutting the contractor's working patterns. The working patterns shall be identical to the Government patterns, except that additional notching, if needed to facilitate manufacture is permitted on the contractor's working pattern. Minor modifications on the working patterns are also permitted where necessary to accommodate the manufacturer's processes and the use of automated equipment provided the modifications do not alter the dimensional, serviceability, or appearance requirements cited in this document or those governed by the Government patterns.

3.5.1 Pattern parts. Component parts shall be cut from the material specified and in accordance with the pattern parts listed in table V.

TABLE V. List of pattern parts.

Material	Piece Name	Cut Parts
Basic material	Leg	2
	Fly	1
Fly lining	Fly backing	1

3.6 Construction. The following specifics will provide garments with uniform appearance, comfort, and durability for use during field duties, combat, and operations other than war. End item construction and appearance shall conform to the requirements and figures provided and the finished measurements cited (see 3.7) to maintain end item configuration.

3.6.1 Seams and stitching. The seams and stitching used shall be consistent, exhibit a uniform appearance, and conform to ASTM D6193 and the requirements for location, stitch, and seam types listed in table VI. Seam allowance shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers, or open seams result. All seams shall start and finish evenly and edges shall not unravel. Edges may be turned-in, turned-under, or serged to prevent raveling. Thread tension shall be maintained so that there is no tight, loose, or unbalanced stitching. The stitches per inch shall be 10-14 for all seams except stitch type 607 which shall be 12-14 stitches per inch.

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TABLE VI. Seams and stitching.

Seaming Area and Operations	Stitch Type	Seam Type
Hem edges of fly opening	605	EFa-2
Set fly to front of drawers	607	FSa-1
Set waistband elastic webbing	403	LSa-2
Close inseams and seat seam	607	FSa-1
Hem ⅝±⅛ inch bottom of legs	605	EFa-2
Place the USMC/size label over center back seam of waistband elastic and stitch 1/16 to 3/16 inch from edges on all four sides. Catch seam tail and top edge of care label in bottom row of USMC/size label stitching	301	SSa-1
Place FROG label on left front of drawers as worn approximately ¾ to 1 inch below waistband	301	LSbj-1

3.6.2 Bartacks. Bartacks for reinforcement shall be placed in the length, quantity, and locations as specified in table VII and as shown in figures 1 and 2.

TABLE VII. Bartacks.

Drawer Bartack Location	Length (Inch)	Quantity per Garment
Hem endings at inseam	⅝	2

3.6.3 Tacking and backstitching. Ends of seams and rows of stitching, when not caught in other seams or stitching, shall be securely backstitched not less than ¼ inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than ½ inch.

3.7 Finished dimensions. The finished drawers shall conform to the measurements listed in table VIII as defined in table IX. All measurements shall be taken with the garment laid flat. Inseam lengths shall not be uneven by more than ½ inch.

TABLE VIII. Finished measurements (inches).

Regular	X-Small	Small	Medium	Large	X-Large	Tolerance
½ waist	10¾	12¼	13¾	15¼	16¾	±½
Inseam	24¼	25½	26¾	28	29½	±½
Long			**Medium**	**Large**	**X-Large**	
½ waist			13¾	15¼	16¾	±½
Inseam			29¾	31	32¼	±½

TABLE IX. Methods of measure.

Measurement	Description
½ waist	On waist elastic from folded edge to folded edge – in a relaxed state
Inseam	Edge of fly insert (at crotch) along inseam to base of hem

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3.8 Toxicity. The drawers shall not present a health hazard and shall show compatibility with prolonged direct skin contact when tested as specified in 4.4.5. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used. When specified (see 6.2), the contractor shall furnish information, which certifies that the finished product is composed of materials, which have been safely used commercially or provide sufficient toxicity data to show compatibility with prolonged, direct skin contact.

3.8.1 Toxicity documents. As specified (see 6.2), all finishes/chemicals used to process the garments shall be identified and accompanied by the appropriate Safety Data Sheets (SDS) information.

3.9 Workmanship. The finished drawers shall be free from loose thread, foreign matter, and irregular defects that can adversely affect usage or durability. The finished drawers shall conform to the quality of product established by this document, be uniform, and be free from defects specified in 4.4.3 and those defects that adversely affect form, fit, or function. Defects shall not exceed the Acceptable Quality Limits (AQLs) (see 6.2) and shall not adversely affect the serviceability, appearance and uniformity of the product.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, in accordance with 3.3.1, shall be examined in accordance with 4.4.1 through 4.4.5 for compliance with design, configuration, workmanship, dimensional requirements, and testing as specified in this document (see 6.2).

4.2.1 First article samples and acceptance criteria. Unless otherwise specified in the contract (see 6.2), first article samples shall be provided. The sample size shall be as specified in the contract (see 6.2). The sample unit shall be one pair of drawers and the lot size shall be expressed in units of drawers. The presence of excessive defects exceeding AQLs (see 6.2) or failure to pass any test shall be cause for rejection of the first article.

4.3 Conformance inspections. Unless otherwise specified in the contract (see 6.2), conformance inspection, in accordance with 3.1.2 shall consist of the examinations and tests specified in 4.4.1 through 4.4.5.

4.3.1 Material and component sampling and acceptance criteria. Unless otherwise specified in subsidiary specifications, referenced documents, or procurement documents (see 6.2), sampling shall be in accordance with table X. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All requirements are applicable to the sample unit. For material testing, the sample unit shall be 5 continuous yards full width of finished cloth for all physical and chemical tests.

TABLE X. Material sampling.

Lot size (Yards)	Sample Size (Sample Unit)
800 or less	2
801 through 22,000	3
22,001 and over	5

4.3.2 End item sampling and acceptance criteria. Unless otherwise specified in the contract (see 6.2) sampling for end item inspection shall be performed conforming to ANSI/ASQ Z1.4. The sample unit shall be one finished pair of drawer and the lot shall be expressed in units of drawers. The presence of excessive defects (see 6.2) or failure of any testing as specified (see 6.2) shall be cause for rejection of the lot.

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4.4 Inspection methods.

4.4.1 Component and material examinations and tests. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of this specification and those referenced specifications and standards unless otherwise excluded, amended, or modified in this specification or applicable procurement documents (see 6.2). In addition to testing provisions specified in referenced documents, materials and components shall be examined and tested in accordance with table XI through table XV. All requirements are applicable to the sample unit. Unless otherwise specified (see 6.2), the materials and components shall be conditioned and tested in accordance with ASTM D1776/D1776M. All requirements are applicable to the sample unit.

TABLE XI. Component and material verification.

Material	Requirement	Verification
Basic material	3.3.1.1	As listed in table XII
Fly lining material	3.3.1.2	As listed in table XIII
Thread	3.3.2.1	A-A-55217 or A-A-55195
Webbing, elastic with integrated draw-cord	3.3.2.2	As listed in table XIV
Braided draw-cord	3.3.2.2.1	As listed in table XV
Combination size, identification, and instruction label	3.3.2.3.3	MIL-DTL-32075
Barcode label/tag	3.3.2.3.4	MIL-DTL-32075

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TABLE XII. Basic material verification.

Material Characteristic	Material Requirement	Test Method
Construction	3.3.1.1	Visual
Fiber content	3.3.1.1	AATCC TM20A
Weight	3.3.1.1	ASTM D3776/D3776M (method C)
Air permeability (minimum)	3.3.1.1	ASTM D737
Burst strength (minimum)	3.3.1.1	ASTM D3787
Colorfastness (minimum)	3.3.1.1	
Laundering (3 cycles)	3.3.1.1	AATCC TM61, option 2A
Crocking	3.3.1.1	
Wet	3.3.1.1	AATCC TM8
Dry	3.3.1.1	AATCC TM8
Light, Xenon	3.3.1.1	AATCC TM16.3, option 3 (85 kJ)
Dimensional stability (3 cycles) (maximum)		AATCC TM135, (1) III, (A) ii
Wales	3.3.1.1	
Courses	3.3.1.1	
Fabric stretch (minimum)		
Wales	3.3.1.1	ASTM D2594 (Loose Fit)
Courses	3.3.1.1	ASTM D2594 (Loose Fit)
Pilling (jersey face) (minimum)		
Initial	3.3.1.1	ASTM D3512/D3512M
After laundering (3 cycles)	3.3.1.1	AATCC TM135, (1) III, (A) ii ASTM D3512/D3512M
Thermal insulation (Clo)		
Initial	3.3.1.1	ISO 11092
After laundering (3 cycles)	3.3.1.1	AATCC TM135, (1) III, (A) ii ISO 11092
Thickness	3.3.1.1	ASTM D1777
Vertical flame – initial (maximum)		
Wales and courses	3.3.1.1	4.4.1.3 ASTM D6413/D6413M
After flame	3.3.1.1	
After glow	3.3.1.1	
Char length	3.3.1.1	
Melt/drip	3.3.1.1	
Vertical flame – after 5 launderings (maximum)		AATCC TM135, 5 cycles, (1) III, (A) ii
Wales and courses	3.3.1.1	4.4.1.3 ASTM D6413/D6413M
After flame	3.3.1.1	
After glow	3.3.1.1	
Char length	3.3.1.1	
Melt/drip	3.3.1.1	
Visual shade matching		
Daylight (D75) (7500±200K)	3.3.1.1	AATCC EP9, option A
Incandescent (2856±200K)	3.3.1.1	AATCC EP9, option A

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TABLE XIII. Fly lining verification.

Material Characteristic	Material Requirement	Test Method
Construction	3.3.1.2	Visual
Fiber content	3.3.1.2	AATCC TM20A
Weight	3.3.1.2	ASTM D3776/D3776M (method C)
Bursting strength (minimum)	3.3.1.2	ASTM D3787
Colorfastness (minimum)		
Laundering (3 cycles)	3.3.1.2	AATCC TM61, option 2A
Crocking	3.3.1.2	
Wet	3.3.1.2	AATCC TM8
Dry	3.3.1.2	AATCC TM8
Dimensional stability (3 cycles) (maximum)	3.3.1.2	
Wales	3.3.1.2	AATCC TM135, 3 cycles, (1) III, A, ii
Courses	3.3.1.2	AATCC TM135, 3 cycles, (1) III, A, ii
Fabric stretch (minimum)		
Wales	3.3.1.2	ASTM D2594
Courses	3.3.1.2	ASTM D2594
Vertical flame – initial (maximum)	3.3.1.2	
Wales and courses	3.3.1.2	ASTM D6413/D6413M
After flame	3.3.1.2	
Char length	3.3.1.2	
Melt/drip	3.3.1.2	
Vertical flame – after 5 launderings (maximum)	3.3.1.2	AATCC TM135, 5 cycles, (1)III, (A) ii
Wales and courses	3.3.1.2	ASTM D6413/D6431M
After flame	3.3.1.2	
Char length	3.3.1.2	
Melt/drip	3.3.1.2	
Visual shade matching		
Daylight (D75) (7500±200K)	3.3.1.2	AATCC EP9, option A
Incandescent (2856±200K)	3.3.1.2	AATCC EP9, option A
Water sorption/wicking		
Back and face		
Initial	3.3.1.2	4.4.1.1
After laundering (20 cycles)	3.3.1.2	AATCC TM135, 20 cycles, (1) III, A ii, 4.4.1.1

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TABLE XIV. Webbing, elastic with integrated draw-cord verification.

Material Characteristic	Material Requirement	Test Method
Fiber content	3.3.2.2	AATCC TM20A
Construction	3.3.2.2	Visual
Width	3.3.2.2	ASTM D3774
Weight (minimum)	3.3.2.2	ASTM D3776/D3776M
Elastic ends (minimum)	3.3.2.2	Visual count
Picks per inch (minimum)	3.3.2.2	Visual count
Warp ends (minimum)	3.3.2.2	Visual count
Load to produce 50% elongation	3.3.2.2	4.4.1.2
Drift, % (maximum)	3.3.2.2	4.4.1.2
Colorfastness		
Laundering (3 cycles)	3.3.2.2	AATCC TM61, option 2A
Crocking		
Wet	3.3.2.2	AATCC TM8
Dry	3.3.2.2	AATCC TM8
Light, Xenon	3.3.2.2	AATCC TM16.3, option 3 (85 kJ)

TABLE XV. Braided draw-cord verification.

Material Characteristic	Material Requirement	Test Method
Fiber content	3.3.2.2.1	AATCC TM20A
Construction	3.3.2.2.1	Visual
Diameter	3.3.2.2.1	Visual/micrometer
Colorfastness		
Laundering (3 cycles)	3.3.2.2.1	AATCC TM61, option 2A
Crocking		
Wet	3.3.2.2.1	AATCC TM8
Dry	3.3.2.2.1	AATCC TM8
Light, Xenon	3.3.2.2.1	AATCC TM16.3, option 3 (85 kJ)

4.4.1.1 Water sorption and wicking. Water sorption and wicking shall be determined using the following procedure:

4.4.1.1.1 Water sorption and wicking test specimens. Fabric specimens shall be conditioned in accordance with ASTM D1776/D1776M and tested in that environment. The specimen size shall be 6 inches by 6 inches; three (3) separate specimens shall be used for each of the face side and back side tests. A fabric shall be considered passing only when tests on both the face side and the back side meet the respective test pass/fail criteria on all individual specimens tested.

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4.4.1.1.2 Face side wicking test. The test specimen shall be laid flat on a glass plate with back side up (i.e., inner or skin surface when used in a garment). One (1) drop of 0.10±0.01 milliliters of distilled water at 70±2 °F shall be placed in the center of the test specimen using a pipettor and a stopwatch/timer immediately started. The test specimen shall then be immediately turned over on the glass plate with test specimen face side up. The diameter of the wicked water area (denoted by a darkened water mark) shall be measured at a total elapsed time of 10 seconds. The specimen shall be considered passing if the diameter of the wicked water area (darkened water mark) is equal to or greater than 1 3/16 inches.

4.4.1.1.3 Back side sorption test. The test specimen shall be laid flat on a glass plate with back side up (i.e., inner or skin surface when used in a garment). One (1) drop of 0.10±0.01 milliliter of distilled water at 70 ±2 °F shall be placed in the center of the test specimen using a pipettor and a stopwatch/timer immediately started. The water (denoted as a darkened water mark) shall be observed and the time for the water mark to disappear (water sorption, denoted as a lightened water mark approximating the shade of the basic material) shall be recorded. The specimen shall be considered passing if the water sorption (disappearance of the darkened water mark) is 10 seconds or less.

4.4.1.2 Waistband elastic tests. Waistband elastic webbing samples shall be samples from garments or constructed to duplicate the configuration on the finished garment. Tensile properties shall be evaluated by conducting tensile tests utilizing equipment which conforms to ASTM D76/D76M. Samples shall be conditioned in accordance with ASTM D1776/D1776M.

4.4.1.2.1 Waistband elongation test. Specimens for the test shall be full-width at least 6 inches long, and shall be marked with a 2-inch gage length so located that when the specimen is inserted in the jaws of a tensile testing machine, having jaws wider than the webbing, the gage length is centrally located between the jaws. The initial distance between the jaws of the machine shall be 3 inches. The jaws shall separate at a rate of 12 inches per minute under no load. The load required to produce 50-percent elongation of the 2-inch gage length shall be noted.

4.4.1.2.2 Waistband drift test. Specimens for the test shall be full-width at least 6 inches long, and shall be marked with a 2-inch gage length so located that when the specimen is inserted in the jaws of a tensile testing machine, having jaws wider than the webbing, the gage length is centrally located between the jaws. The initial distance between the jaws of the machine shall be 3 inches. The jaws shall separate at a rate of 12 inches per minute until 50-percent elongation is reached. Jaw separation shall be stopped and the load shall be recorded 15 seconds after the instant of stopping. After 4 hours, the load shall be recorded again. The drift shall be determined as follows:

$$\text{Drift, percent} = \frac{a - b}{a} \times 100$$

Where: a = load reading at start of test period

b = load reading at end of test period.

4.4.1.3 Modified vertical flame test. The vertical flame test shall be conducted on samples of the basic material with an aramid cloth, which meets the requirements in table XVI. Both materials shall be cut in the same direction for each test. The basic material cloth shall be placed in the sample holder, face side facing up. The bottom edge (toward the flame) shall be folded back ½ inch. The back side of the fabric shall now be facing the flame. The aramid cloth shall be placed along the edge of the basic material cloth and secured in the sample holder. The vertical flame test shall be performed in accordance with ASTM D6413/D6413M.

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TABLE XVI. Aramid cloth requirements.

Material Characteristic	Material Requirement	Test Method
Construction	Plain weave	Visual
Fiber content	95% Meta-aramid/5% Para-aramid	AATCC TM20A
Fiber denier	1.5 denier per filament	ASTM D1577
Fiber staple length	1½ to 2 inches	ASTM D5103
Yarn type		
Warp	2 ply spun	Visual
Filling	2 ply spun	Visual
Weight		ASTM D3776/D3376M
Breaking strength		ASTM D5034
Warp	125 pounds	
Filling	125 pounds	
Char temperature (minimum)	675 °F	Thermogravimetric analysis
Dimensional stability (3 cycles) (maximum)		AATCC TM135, 3 cycles, (1) III, (A) ii
Warp	3.5%	
Filing	3.5%	
Vertical flame (initial) (maximum)		
After flame	2.0 seconds	ASTM D6413/D6413M
After glow	0 seconds	
Char length	5.0 inches	
Melt/drip	No melt/no drip	
Vertical flame (after 5 launderings) (maximum)		AATCC TM135, 5 cycles, (1) III, (A) ii
After flame	2.0 seconds	ASTM D6413/D6413M
After glow	0 seconds	
Char length	5.0 inches	
Melt/drip	No melt/no drip	

4.4.2 In-process examination. Visual and dimensional examinations shall be made at any point or during any phase of the manufacturing process to determine whether construction details, which cannot be examined in the finished product, are in accordance with the requirements specified in section 3. Materials and components which contain defects and damages, as defined in the visual examination of 4.4.3, shall be removed from production. Any in-process nonconformance remaining in the finished drawers shall be classified as a defect in accordance with the visual examination of 4.4.3.

4.4.3 Visual examination. The drawers shall be visually examined shall be examined for defects in color, shade, design, material, construction, workmanship, and markings, and the defects shall be classified in accordance with the list of defects specified in table XVII. The inspection conditions and the AQLs shall be as specified (see 6.2)

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TABLE XVII. Material and end item visual examination.

Examination	Defect	Classification	
		Major	Minor
Workmanship	Component part omitted, distorted, full, tight, or twisted; any part of end item caught in any unrelated stitching; the edge of any component part required to be forced out having folds of more than ⅛ inch.	101	
Material	Hole, cut, tear, smash, burn, drill hole, run, needle chew, visible mend, thin place, or misweave affecting appearance or serviceability.	102	
Shade	a. Shade variation within a part or between parts. Thread color not as specified.	103	
	b. Dye streak; color not as specified	104	
Seams and stitching	a. Seams puckered, distorted, pleated, wavy, or twisted; edge or raised stitching sewn too close to edge resulting in damage to cloth.		201
	b. Irregular or open seam; raw edge affecting appearance or serviceability	105	
	c. Seam allowance not as specified		202
	d. Loose or tight stitch tension		203
Bartacks	Bartacks missing, insecure, misplaced, not specified size, stitches loose or broken.		204
Cleanliness	Spot, stain, excessive thread ends not trimmed or removed, odor		205
Waistband and integrated draw-cord	a. Any draw-cord caught in tunnel or stitching restricting use of draw-cord	106	
	b. Any draw-cord omitted or not positioned correctly	107	
	c. Draw-cord not exiting center front of waistband	108	
	d. Draw-cord not knotted at exit		206
	e. Any draw-cord insufficient in length for function	109	
Labels	a. Label omitted, incorrect type, size or configuration; illegible; inaccurate; or not attached as specified	110	
	b. Barcodes omitted or not readable by scanner; human-readable interpretation (HRI) omitted or illegible		207
	c. Barcode not visible on folded, packaged item		208
	d. Barcode attachment causes damage to the item	111	
Packaging	Any pair of drawers not packaged in accordance with the contract or purchase order		209

4.4.4 Dimensional examination. The drawers shall be examined for conformance to the dimensions specified in table VIII as defined in table IX. Any measurement deviating from the dimensions and tolerances specified shall be scored as a measurement defect. The inspection conditions and the AQLS shall be as specified (see 6.2).

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4.4.5 Toxicity assessment. An acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of the studies indicate the drawers are not a sensitizer or irritant, a human Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (Hayes' Principles and Methods of Toxicology and Contact Allergy: Predictive Testing in Humans). If the toxicity requirement (see 3.8) can be demonstrated with historical data, then toxicity testing may not be required. At a minimum, toxicity data should include results from a primary dermal irritation study in laboratory animals and a repeated insult human patch test (Modified Draize Procedure). The latter must be conducted under the supervision of a qualified dermatologist using at least 100 free-living individuals.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The drawers are part of the underwear set intended for use with the cold weather clothing system worn by personnel of the USMC.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Sizes and lengths required (see 1.1).

c. National stock number (NSNs).

d. When first article is required (see 3.1.1, 4.2, and 6.3).

e. When conformance inspection is required (see 3.1.2 and 4.3).

f. Applicable Government patterns and reference standards, including revisions (see 3.3.1.1, 3.3.1.2, and 3.5).

g. Barcode/label tag location (see 3.3.2.3.4).

h. Requirement for the submittal of information which certifies that the finished product is composed of materials which have been safely used commercially; or sufficient toxicity data to show compatibility with prolonged, direct skin contact (see 3.8).

i. Requirement for an SDS and the identification of all finishes/chemicals used to process the garment (see 3.8.1).

j. AQLs (see 3.9 and 4.4.4).

k. First article sampling and acceptance criteria (see 4.2.1).

l. Component and material sampling and acceptance criteria (see 4.3.1).

m. End item sampling and acceptance criteria (see 4.3.2).

n. Component and material examination and tests (see 4.4.1).

o. Inspection conditions (see4.4.3 and 4.4.4).

p. Packaging requirements (see 5.1).

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6.3 First article. When first article inspection (see 6.2) is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2.1. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for pending contract. Bidders should not alternate submit bids unless specifically requested to do so in the solicitation.

6.4 Suggested sources.

a. Basic material and fly lining:

(1) Polartec, LLC, 46 Stafford St., Lawrence, MA 01841, (978) 685-6341

b. Woven labels:

(1) Bell Label, 777 Main St., Lewiston, ME 04240, (207) 784-2961

(2) Ibis Tek, LLC, 912 Pittsburgh Rd., Butler, PA 16002, (724) 586-6005

c. Webbing, elastic with draw-cord:

(1) Northeast Knitting, Inc., 179 Conant St., Pawtucket, RI 02860, (401) 727-0500

6.5 NSNs. For informational purposes only, an NSN that corresponds to this specification is listed in table XVIII. Table XVIII is not intended to be a comprehensive list of all NSNs associated with this specification.

TABLE XVIII. NSN.

NSN	Size
8415015553809	Medium Regular

6.6 Supersession data. This document supersedes MC/PS 04 2015 SYSCOM, dated 15 March 2015, in its entirety.

6.7 Subject term (key word) listing.

Grid fleece

Level 2

Marine Corps

Underwear

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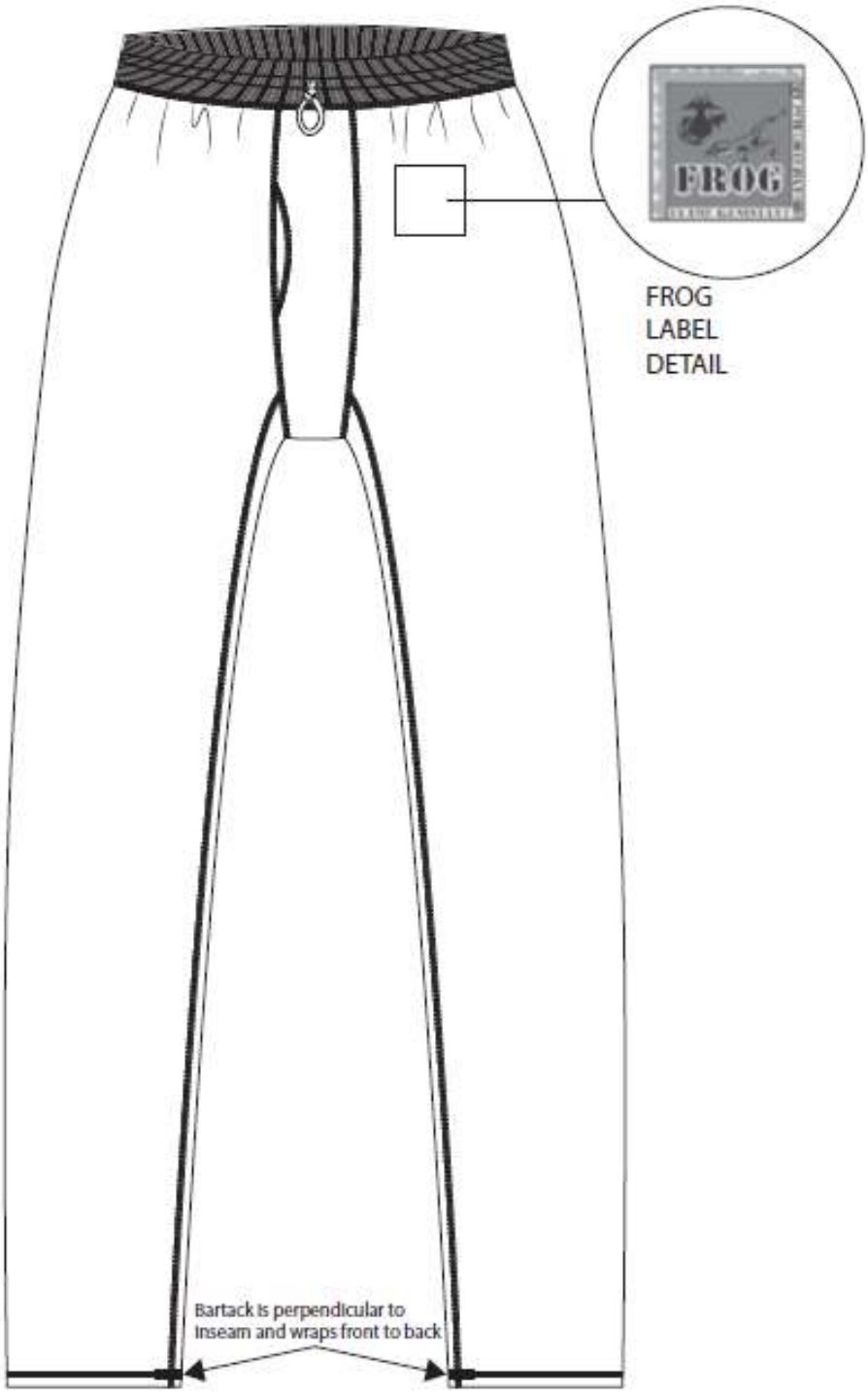


FIGURE 1. Front view.

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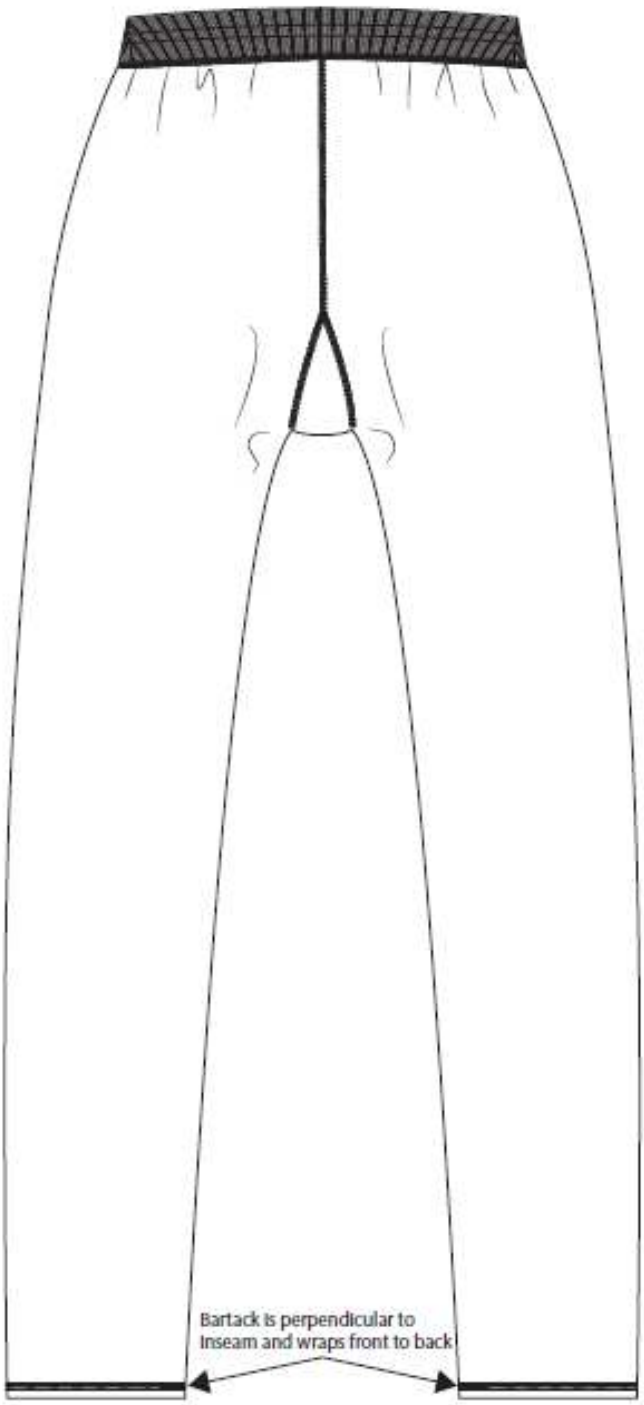


FIGURE 2. Back view.

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CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – MC
Air Force – 11

Review Activities
Army – AV, MI
Navy – CG1, NU
Air Force – 03, 70
DLA – CT

Civil agency
GSA – FAS

Preparing activity:
Navy – MC
(Project 8415-2018-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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